

# Assembly Line

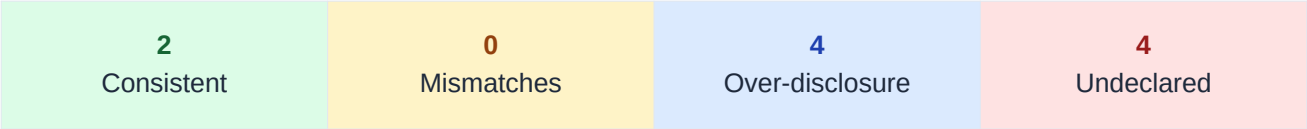
com.olympus.assemblyline

|            |               |
|------------|---------------|
| Developer  | Olympus       |
| Genre      | GAME_STRATEGY |
| Installs   | 1,000,000+    |
| Audit Date | 2026-04-26    |

Overall Risk Score

0.52

MEDIUM



Research prototype · Ongoing research | Policy: [http://olympus.coolpage.biz/privacy\\_policy.html](http://olympus.coolpage.biz/privacy_policy.html)

## Executive Summary

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**Assembly Line** declares 0 data type(s) collected and 1 shared in its Play Data Safety label, across 10 data type(s) analyzed. Our heterogeneous GNN audit model identifies **4 potential undeclared collection(s)** — data types implied by embedded SDK signals but absent from both label and policy — and **0 policy-vs-label mismatch(es)**. Overall risk score: **0.52 (MEDIUM)**.

### Top discrepancies by risk:

- **installed\_apps**: Undeclared Collection (confidence 1.00)
- **crash\_logs**: Undeclared Collection (confidence 1.00)
- **diagnostics**: Undeclared Collection (confidence 1.00)

### Class definitions:

**CONSISTENT**: Label, policy, and SDK signals agree.

**POLICY\_LABEL\_MISMATCH**: Label discloses data not mentioned in policy.

**OVER\_DISCLOSURE**: Policy mentions data not declared in label.

**UNDECLARED\_COLLECTION**: SDK signals collection undisclosed in label and policy.

## Discrepancy Table

Sorted by risk class (undeclared collection first). Y/N: label=data-safety label, policy=policy text, sdk=SDK signal.

| Data Type            | Class                 | Conf. | Evidence                            |
|----------------------|-----------------------|-------|-------------------------------------|
| installed_apps       | Undeclared Collection | 1.00  | label:N policy:N sdk:Y (1, 2, 4 +1) |
| crash_logs           | Undeclared Collection | 1.00  | label:N policy:N sdk:Y (1, 2, 4 +1) |
| diagnostics          | Undeclared Collection | 1.00  | label:N policy:N sdk:Y (1, 2, 4 +1) |
| approximate_location | Undeclared Collection | 0.97  | label:N policy:N sdk:Y (1, 2, 4 +1) |
| phone_number         | Over Disclosure       | 1.00  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| date_of_birth        | Over Disclosure       | 1.00  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| files_and_docs       | Over Disclosure       | 1.00  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| app_interactions     | Over Disclosure       | 0.97  | label:N policy:Y sdk:N (1, 2, 4 +1) |
| device_id            | Consistent            | 1.00  | label:N policy:Y sdk:Y (1, 2, 4 +1) |
| advertising_id       | Consistent            | 1.00  | label:N policy:Y sdk:Y (1, 2, 4 +1) |

## Evidence Trails

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Detailed evidence for UNDECLARED\_COLLECTION or confidence > 0.85 non-CONSISTENT findings.

### Discrepancy: installed\_apps — Undeclared Collection (confidence 1.00)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'installed\_apps' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: crash\_logs — Undeclared Collection (confidence 1.00)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'crash\_logs' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: diagnostics — Undeclared Collection (confidence 1.00)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'diagnostics' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

### Discrepancy: approximate\_location — Undeclared Collection (confidence 0.97)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **No**

SDK collection signals: **Yes** — 1, 2, 4, 35

Recommended action: Add 'approximate\_location' to the Play Data Safety label under 'Data Collected' and update the privacy policy to disclose this collection.

#### Discrepancy: phone\_number — Over Disclosure (confidence 1.00)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'phone\_number' is genuinely collected. If yes, add to label; if no, remove from policy.

#### Discrepancy: date\_of\_birth — Over Disclosure (confidence 1.00)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'date\_of\_birth' is genuinely collected. If yes, add to label; if no, remove from policy.

#### Discrepancy: files\_and\_docs — Over Disclosure (confidence 1.00)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'files\_and\_docs' is genuinely collected. If yes, add to label; if no, remove from policy.

#### Discrepancy: app\_interactions — Over Disclosure (confidence 0.97)

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Label declares collection: **No**

Label declares sharing: **No**

Policy text mentions: **Yes**

SDK collection signals: **No** — 1, 2, 4, 35

Recommended action: Review whether 'app\_interactions' is genuinely collected. If yes, add to label; if no, remove from policy.

## Methodology

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PolicyGraphAudit-RT represents each Android app as a tri-partite heterogeneous graph connecting three layers: **Privacy Policy** text (OPP-115 segment classification), **Play Data Safety Labels** (declared data types and purposes), and **Runtime Evidence** (inferred SDK trackers via Exodus Privacy). A 2-layer HeteroConv R-GCN encodes all three layers jointly; an MLP classifier head predicts discrepancy classes for each (App, DataType) pair.

The model was trained under an edge-masking protocol (30% of label-determining edges removed) to prevent circularity. It achieves **macro F1 = 0.956** on 39 held-out apps (UNDECLARED\_COLLECTION F1 = 0.974), trained on 252 apps and 3,202 labeled pairs. Model card: [https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5\\_model\\_card.md](https://github.com/PolicyGraphAudit-RT/reports/blob/main/reports/m5_model_card.md).

Known limitations: (1) Privacy labels are from a 2022 Play Store snapshot. (2) SDK presence is inferred via category-level priors, not measured from APK bytecode. (3) Policy classifier uses MiniLM + logistic regression (OPP-115 macro F1 = 0.70). (4) Labels are rule-derived weak supervision, not human-annotated. (5) English-language Android apps only.

## Disclaimer

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This report is produced by a research prototype and is provided for informational and research purposes only. It does not constitute legal advice or a regulatory determination. Findings are based on automated weak-supervision labels from publicly available data and have not been reviewed by a privacy legal professional. Research prototype · Ongoing research.